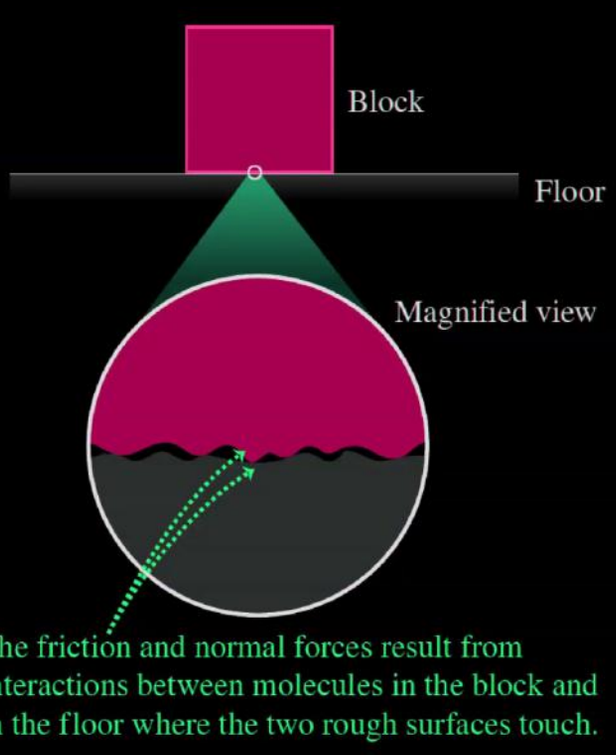
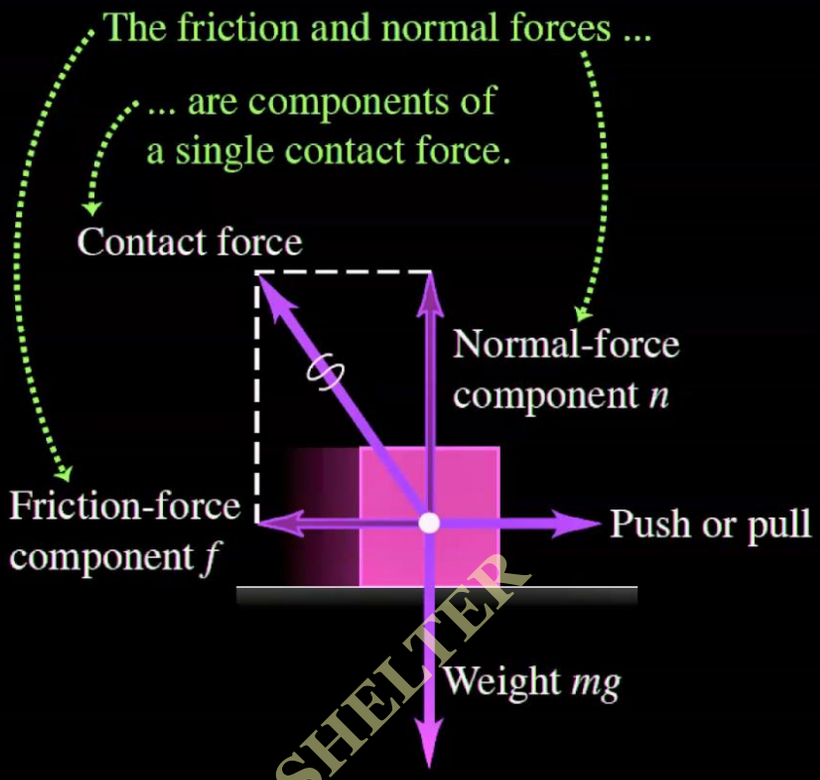
FRICTION:

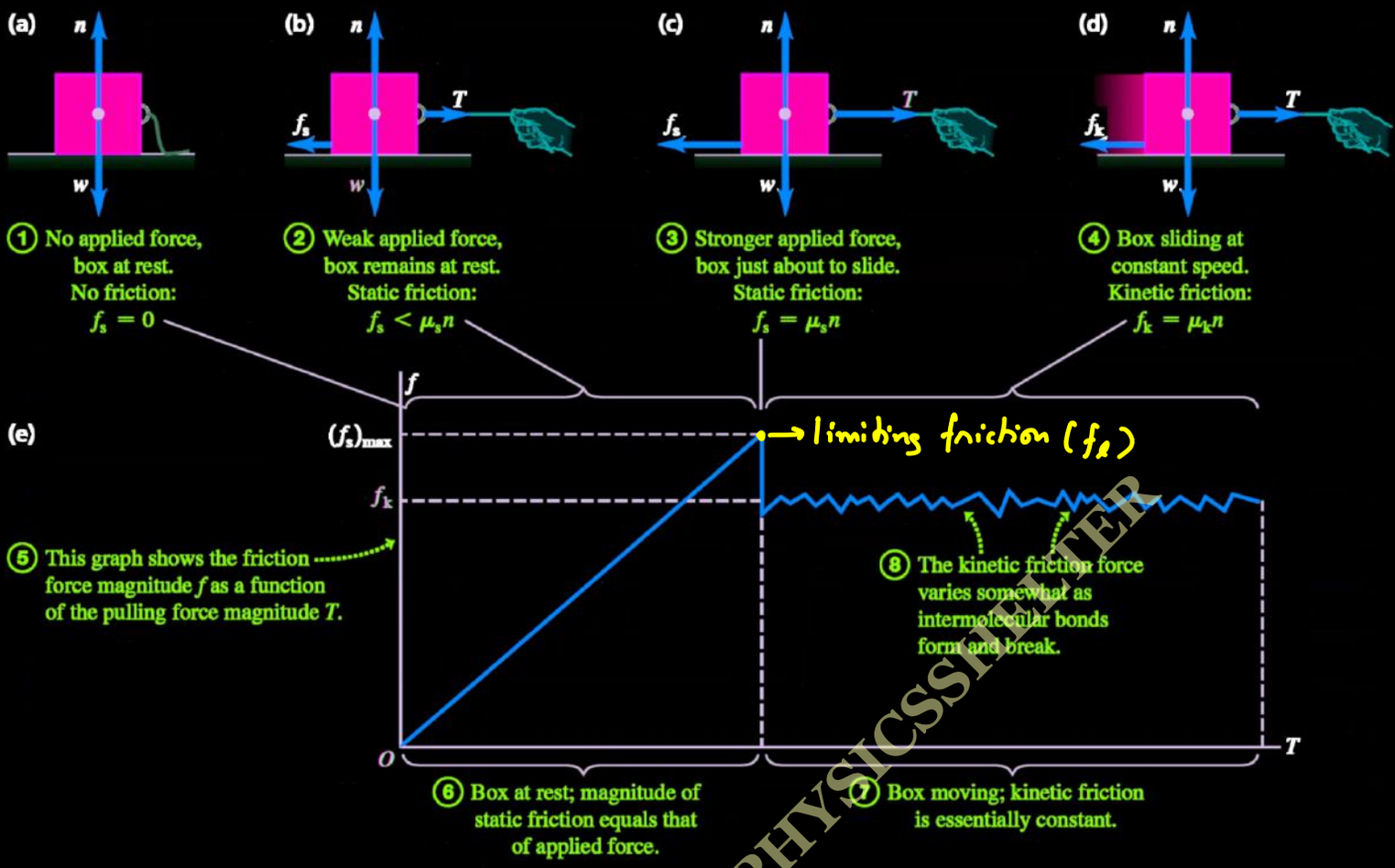




Consider a block being pushed while being placed on a rough surface. We find that if the pushing force is not large enough, the block does not move, which means an opposing force must come in to existence. This opposing force comes due to bond formation between the two contacting surfaces (known as cold welds). The cold welds may be visualized as miniature springs. As we increase the pushing force, the cold welds bear more and more load and if the force exceeds a certain threshold, these bonds break and the motion starts. When the motion starts, these bonds still keep forming and breaking and the force offered by these cold weld bonds parallel to the surface is called friction. When the relative slipping between surfaces has not started, the friction is called static friction and when the slipping is going on the friction is called kinetic friction. Since bonds are not that effectively formed when slipping is going on, the kinetic friction is smaller than limiting value of static friction (i.e static friction when the block is at the verge of slipping).

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PLOT OF PULLING FORCE VS FRICTION:



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The figure above shows plot of applied force vs frictional force for a block being pulled / pushed by a force parallel to the table. Until slipping happens, the applied force is exactly balanced by the frictional force and after that the frictional force becomes nearly constant. The exact shape of the graph is governed by Normal reaction and coefficients of static and kinetic frictions (μ_s and μ_k) about which we will learn in the following section.

LAWS OF DRY FRICTION:

① The limiting friction (f_l) and kinetic friction (f_k) are directly proportional to the normal reaction between contacting surfaces with proportionality constants being μ_s and μ_k respectively, thus:

$$f_l = \mu_s N \qquad f_k = \mu_k N$$

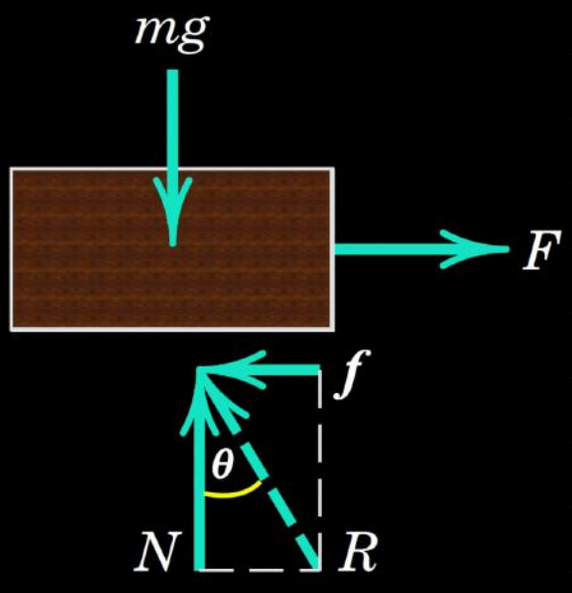
For most surfaces the difference between the kinetic and static coefficient (μ_s) of friction is very small and if not specified, we may assume them to be equal for purpose of solving the problems. (Though $\mu_k \leq \mu_s$ because bonds formed while sliding are not that effective). The μ_s and μ_k depend on material and surface finish.

(2) μ_s and μ_k do not depend on Area of contact (i.e. for a given normal reaction, the limiting and kinetic friction have a fixed value irrespective of area of contact.

→ Caution: Don't just mechanically put friction $f = \mu_s N$, if there is no slipping friction could be anywhere between 0 and $\mu_s N$.

Angles Of Friction And Cones Of Friction:

Consider a block being pulled by a force F as shown in the figure. As the force F is increased, θ increases and reaches a maximum value ϕ_s such that:



$\tan\phi_s = \frac{\mu_s N}{N} = \mu_s$. This angle ϕ_s is called angle of static friction thus:

$$\boxed{\tan\phi_s = \mu_s}$$

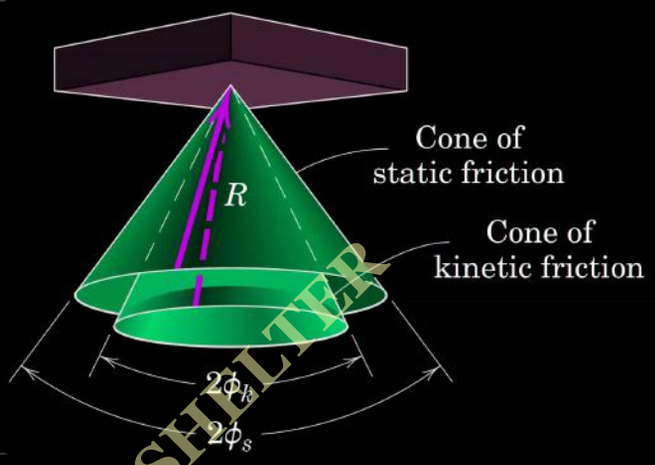
Now when the pulling force is further increased, the value of friction becomes constant and θ attains a value ϕ_k such that:

$$\tan\phi_k = \frac{\mu_k N}{N} = \mu_k \quad \text{thus} \quad \boxed{\tan\phi_k = \mu_k}$$

the angle ϕ_k is called angle of kinetic friction. (Depending on pull direction it could be formed in any direction with vertical).

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The friction angle defines limiting direction of total reaction force R between two contacting surface. If the motion is impending, the resultant contact force must lie on surface of a right circular cone of vertex angle $2\phi_s$. Similarly if slipping has started, the resultant contact force will be somewhere on a right circular cone of apex angle $2\phi_k$. If the slipping has not started, the resultant contact force must lie within the cone of vertex angle $2\phi_s$. These cones are called cones of friction (Cone of static friction and cone of kinetic friction).



→ The idea of cone of friction is sometimes very useful in quickly solving problems involving limiting equilibrium.

Three Types Of Friction Problems:

1. **Condition Of Impending Motion Is Known To Exist:** Here since we know that contacting surfaces are about to slip we can straight away write frictional force $f = \mu_s N$ (since impending motion means same thing as achieving limiting friction condition.

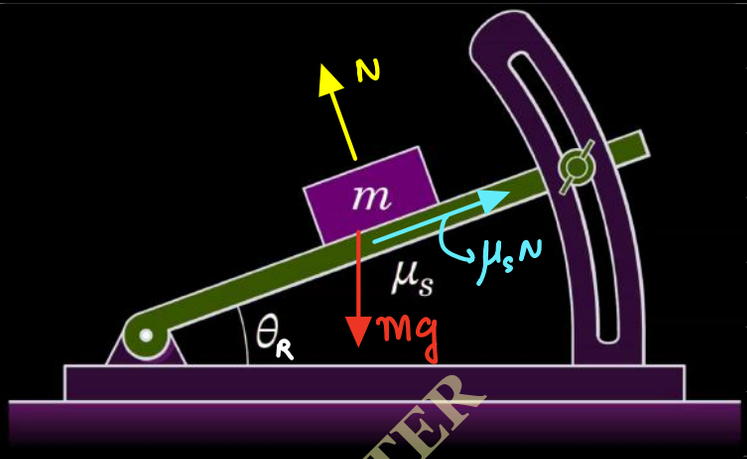
2. **Relative Motion Is Known To Exist:** In this category of problems since slipping is happening, the friction is kinetic and we can straight away write frictional force $f = f_k = \mu_k N$ and apply laws of motion to find accelerations etc.

3. **Neither Impending Motion Nor Relative Motion Is Known A Priori:**
In this type of systems, we are not sure whether friction is kinetic or static beforehand and we have to start solving with certain assumption and later on verify the assumption.

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There are two ways of dealing with this type of problems about which we will learn in a following section.

Angle Of Repose:



Suppose we place a block on an incline and keep increasing the inclination gradually then at a certain angle the block will slip. The angle at which it happens is called angle of repose. We can use the angle of repose to quickly determine the coefficient of friction between a pair of surfaces. This is a case of impending motion so we can directly put $f = \mu_s N$. Now apply force balance equations along the incline and perpendicular to it:

$$mg\cos\theta_R = N \quad (1)$$

$$mg\sin\theta_R = \mu_s N \quad (2)$$

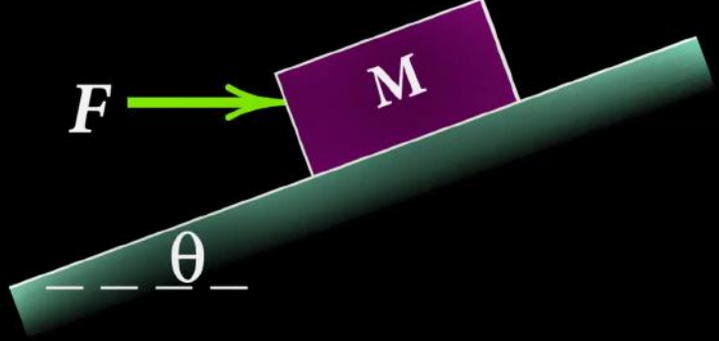
Now dividing (2)/(1):

$$\tan\theta_R = \mu_s$$ thus static friction coefficient is:

$$\boxed{\mu_s = \tan\theta_R}$$

→ Recall that earlier we showed $\mu_s = \tan\phi_s$ where ϕ_s was angle of static friction. Thus angle of repose is equal to the angle of static friction.

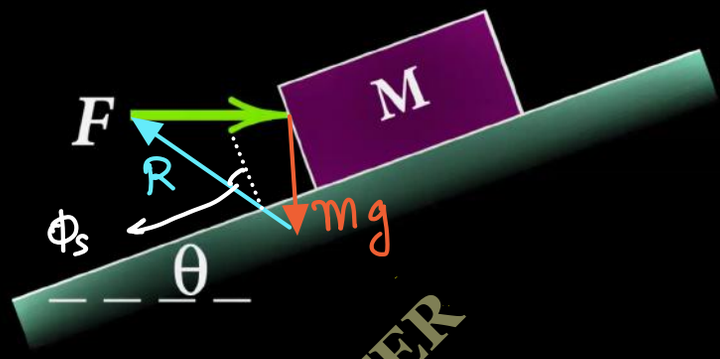
Example: Find the minimum force required to push the block up the plane if the coefficient of static friction is μ. Incline is at rest and initially block is also at rest.



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Solution: This is a case of impending motion. We could solve this problem using force balance however we will prefer a geometrical solution using concept of cone of friction.

With limiting friction, resultant contact force will act at an angle ϕ_s to the normal of the incline thus giving the shown closed triangle as per triangle law of addition.



Now using Sine law :

$$\frac{F}{\sin(\theta+\phi_s)} = \frac{R}{\sin 90^\circ} = \frac{mg}{\cos(\theta+\phi_s)} \Rightarrow F = mg\tan(\theta+\phi_s)$$

Thus $F = mg\left[\frac{\tan\theta+\tan\phi_s}{1-\tan\theta\tan\phi_s}\right] = mg\left[\frac{\mu+\tan\theta}{1-\mu\tan\theta}\right]$ Ans.

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EXAMPLE: MORIN: 2·4:

Keeping a book up *

A book of mass M is positioned against a vertical wall. The coefficient of friction between the book and the wall is μ. You wish to keep the book from falling by pushing on it with a force F applied at an angle θ with respect to the horizontal ($-\pi/2 < \theta < \pi/2$), as shown in Fig. 2.10.

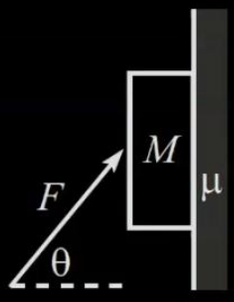


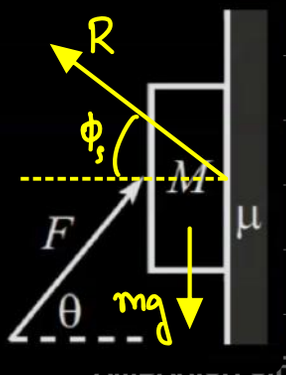
Fig. 2.10

(a) For a given θ, what is the minimum F required?

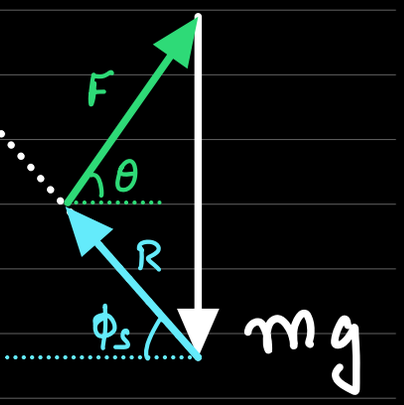
(b) For what θ is this minimum F the smallest? What is the corresponding minimum F?

(c) What is the limiting value of θ, below which there does not exist an F that keeps the book up?

Solution: Once again we have a problem of impending motion and we will prefer a geometrical solution instead of one requiring calculus. In the limiting equilibrium, the book will be at the verge of slipping down and f will be accordingly acting at an angle ϕ_s as shown.



Now we draw the vector triangle as shown in adjacent figure. Now by sine law:



$$\frac{F}{\cos\phi_s} = \frac{R}{\cos\theta} = \frac{mg}{\sin(\theta+\phi_s)}$$

$$\Rightarrow F = \frac{mg\cos\phi_s}{\sin(\theta+\phi_s)} = \frac{mg\cos\phi_s}{\sin\theta\cos\phi_s + \cos\theta\sin\phi_s} \quad (1)$$

$$\Rightarrow F = \frac{mg}{\sin\theta + \mu\cos\theta} \quad \text{Ans. (a).}$$

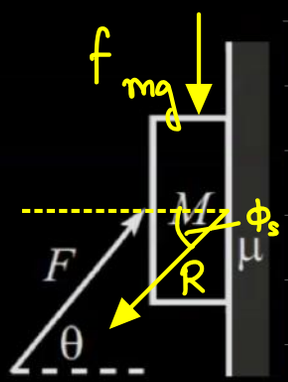
Now this will be minimum when $(\theta+\phi_s)$ is maximum (see (1) middle). Thus for minimum value of F, $\theta+\phi_s = 90°$ or $\theta = 90° - \phi_s$.

i.e $\theta = 90° - \tan^{-1}\mu$., (corresponding $F_{min} = mg\cos\phi_s = \frac{mg}{\sqrt{1+\mu^2}}$ Ans. (b)

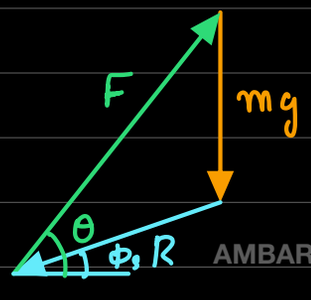
Now for equilibrium to be possible, F should be a positive value, otherwise, Normal reaction will vanish and so also friction. Thus $\sin\theta + \mu\cos\theta > 0$ i.e $\tan\theta + \mu > 0$, thus $\theta > \tan^{-1}(-\mu)$

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Think: What will be the maximum allowable force for a given θ in the previous problem?



Solution: If F is large enough, the block will start sliding upwards, thus R will be acting downwards at an angle ϕ_s with normal. This gives us the vector triangle as shown in the next figure. Thus:



$$\frac{mg}{\sin(\theta-\phi_s)} = \frac{R}{\cos\theta} = \frac{F}{\sin(90°+\phi)}$$

$$\Rightarrow F = \frac{mg\cos\phi}{\sin(\theta-\phi_s)}, \text{ Here } \theta \text{ should be } > \text{than } \phi_s.$$

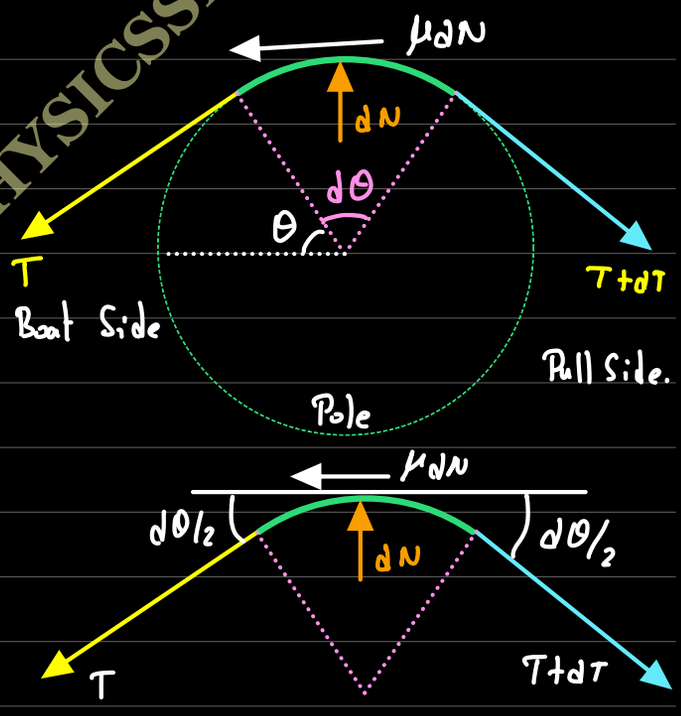
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If $\theta < \phi_s$, maximum force could even be infinite, because, ϕ_s could then be replaced by some ϕ within cone of friction which would make denominator zero and hence equilibrium possible even with infinite force.

Tension Ratio For A Rope Around A Pole With Impending Slip:

Example: (Morin):

Example (Rope wrapped around a pole): A rope wraps an angle θ around a pole. You grab one end and pull with a tension T_0. The other end is attached to a large object, say, a boat. If the coefficient of static friction between the rope and the pole is μ, what is the largest force the rope can exert on the boat, if the rope is not to slip around the pole?



Solution: Consider the FBD of a small element of the rope subtending an angle $d\theta$ at the centre of the rope. Let dN be the normal reaction on it and let tensions on its either sides be T and $T+dT$

Now force balance along tangential direction gives:

$$T\cos(d\theta/2) + \mu dN = (T+dT)\cos(d\theta/2)$$

But for very small $d\theta$, $\cos d\theta \approx 1$

Thus:

$$\mu dN = dT \quad (1)$$

Now using radial force balance:

$$T\sin\left(\frac{d\theta}{2}\right) + (T+dT)\sin\left(\frac{d\theta}{2}\right) = dN$$

But $\sin(d\theta/2) \simeq d\theta/2$ and $dT \times d\theta$ is very small second order term. Neglecting second order term and using small angle approximation:

$$Td\theta = dN \quad \text{or} \quad dN = Td\theta \quad (2)$$

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Now ② / ① gives $\mu = \frac{dT}{T d\theta} \Rightarrow \frac{dT}{T} = \mu d\theta$, now let tension on the boat side be T, now integrating in appropriate limits:

$$\int_{T}^{T_0} \frac{dT}{T} = \int_{0}^{\theta} \mu d\theta \Rightarrow \ln \frac{T_0}{T} = \mu\theta$$

or $\boxed{T = T_0 e^{-\mu\theta}}$

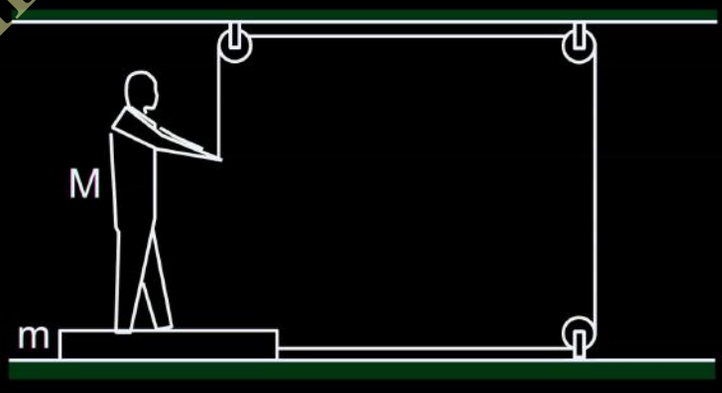
→ Tension keeps declining exponentially with angle. The above formula could be used to find tension at an intermediate point also.

→ If the slipping has started we could still use the same formula if the string were massless or pulling happens at constant speed except that we will need to substitute kinetic friction coefficient.

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EXAMPLE :

In the shown system, coefficient of friction between the floor and the board is μ. What is the maximum force that man can apply on the rope so that board does not slip on the floor.



Solution: A combined FBD of the man + board is made considering impending motion. Here T is the force with which man pulls the rope.

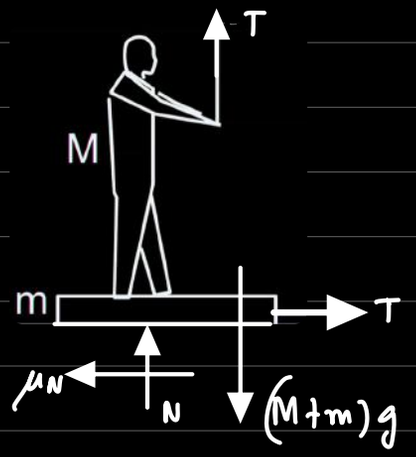
Now by vertical force balance:

$$T + N = (M+m)g \quad ①$$

By horizontal force balance:

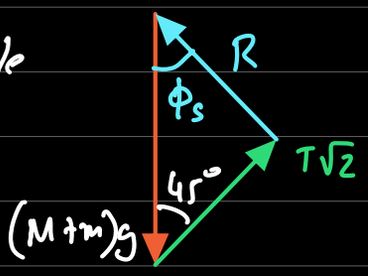
$$T = \mu N \quad ②$$, using ② in ①:

$$T[1 + 1/\mu] = (M+m)g \Rightarrow T = \frac{\mu (M+m) g}{1+\mu}$$



Alternative geometrical solution:

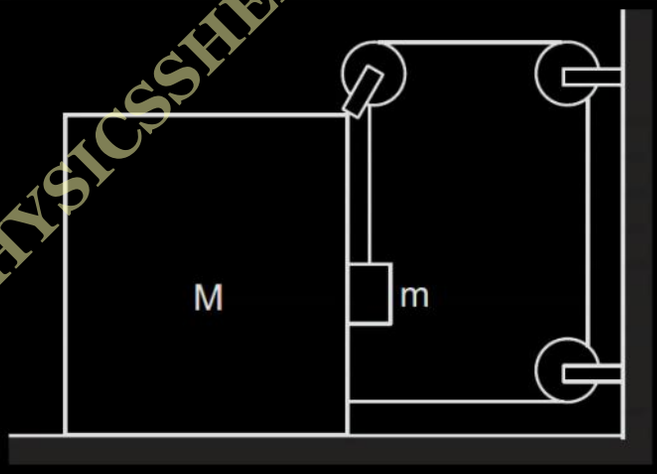
We can combine the two tensions in to $T\sqrt{2}$ at angle $45°$ with vertical. Thus we get the shown vector triangle. Now by sine law:



$$\frac{(M+m)g}{\sin(45°+\phi_s)} = \frac{T\sqrt{2}}{\sin\phi_s} \Rightarrow T = \frac{(M+m)g}{\sqrt{2}} \frac{\sin\phi_s}{\sin(45°+\phi_s)}$$

$$\Rightarrow T = \frac{(M+m)g}{\sqrt{2}} \left[\frac{\sin\phi_s}{\sin\phi_s \times \frac{1}{\sqrt{2}} + \cos\phi_s \times \frac{1}{\sqrt{2}}} \right] = \frac{(M+m)g\tan\phi_s}{\tan\phi_s + 1} = \frac{\mu}{\mu+1}(M+m)g \quad \text{Ans.}$$

EXAMPLE: Assuming that slipping does occur, find the acceleration of M in the adjacent figure. Coefficient of friction at ground is μ_2 and between blocks it is μ_1.



Solution: Considering combined system and apply $\Sigma m_i \vec{a}_i = \Sigma \vec{F}_{i\,ext}$,

In x direction:

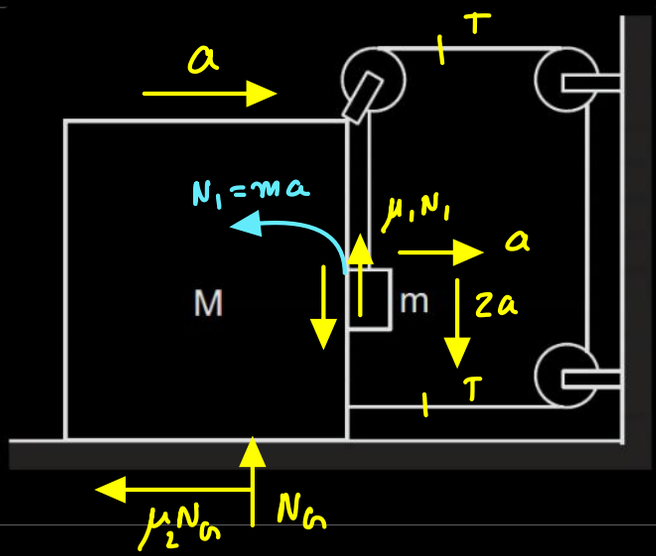
$$2T - \mu_2 N_G = (M+m)a \quad (1)$$

In vertical direction:

$$(M+m)g - N_G = 2ma \quad (2)$$

$N_1 = ma$ ($\because$ FBD of m) accordingly for block m in vertical direction:

$$mg - T - \mu_1 ma = 2ma \quad (3)$$



Solving (1), (2) & (3):

$$a = \frac{[2m - \mu_2(M+m)]g}{M + m[5 + 2(\mu_1 - \mu_2)]} \quad \text{Ans.}$$

→ If we did not know whether slipping was happening or not, we could assume slipping to calculate a and then a

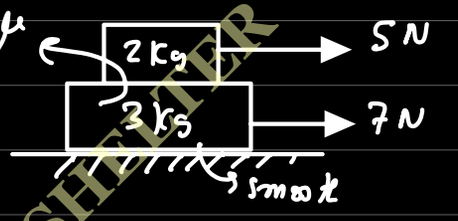
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must be greater than zero with assumed direction. If it is negative, our assumption would be wrong.

How To Decide Direction Of Static Friction:

Static friction always acts opposite to the tendency of relative motion. To find the direction of static friction, think what would happen if surface were frictionless. If there is slipping if the surface were assumed frictionless, then friction would act opposite to the relative slipping direction.

For instance suppose μ is large enough to move the two blocks together in adjacent figure, then if friction were absent, $a_2 = 5/2 = 2.5\ m/s^2$ → and $a_3 = 7/3 \approx 2.33\ m/s^2$ → Thus 2 kg is relatively slipping to the right if surface were assumed smooth ∴ friction on 2 kg must act leftwards and that on 3 kg must act rightwards.



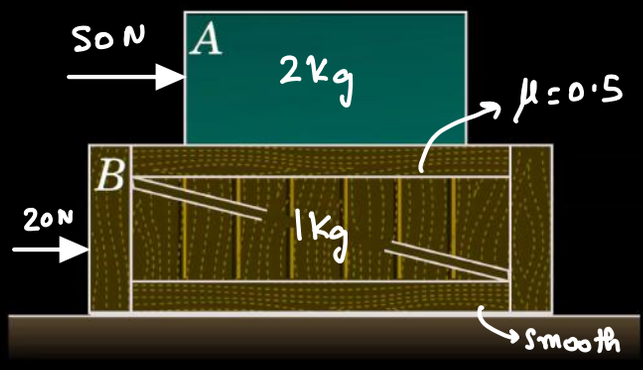
How To Decide Whether Friction Is Kinetic Or Static:

Method I: Assuming Static Friction:

1. Assume friction to be static
2. Find accelerations of blocks under this assumption
3. Find required frictional force for causing the accelerations obtained
4. Compare the required friction with limiting friction. If required friction is more than limiting friction, then our assumption has failed and we need to solve again assuming kinetic friction otherwise if required friction is less than limiting friction, our assumption is correct and we have the correct solution with us.

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EXAMPLE: For the shown situation, find acceleration of the blocks and the frictional force between them.

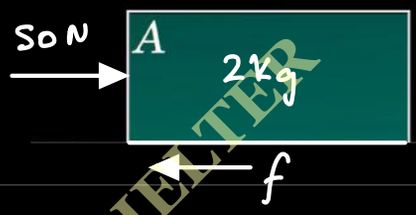


Solution: Assuming static friction,

$a_1 = a_2 = (50+20)/(2+1) = 70/3\ ms^{-2}$.

Now considering the FBD of upper block alone, $50 - f = 2 \times 70/3$

$\Rightarrow f = 50 - 140/3 = 10/3$ N.



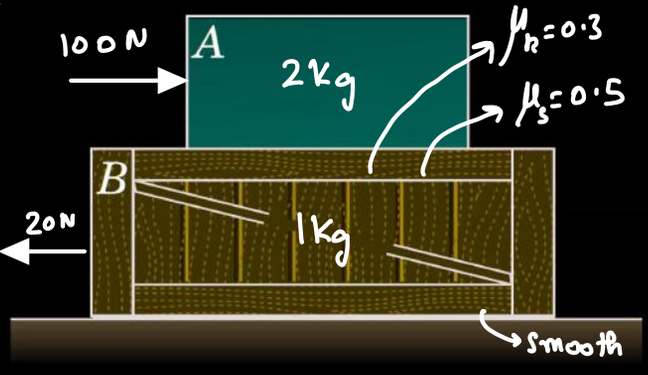
Now limiting friction, $f_l = 20 \times 0.5 = 10$ N

$\because f < f_l$ thus our assumption is correct. Thus:

$a_1 = a_2 = 70/3\ ms^{-2}$ → and $f = 10/3$ N Ans.

→ Under assumption of static friction, it does not matter even if you apply friction in wrong direction. Upon solving if it comes negative, it simply means that it is in opposite direction with same magnitude, then comparison must be made between $|f|$ and f_s.

EXAMPLE: For system in adjacent figure, find accelerations of the blocks and friction between them.



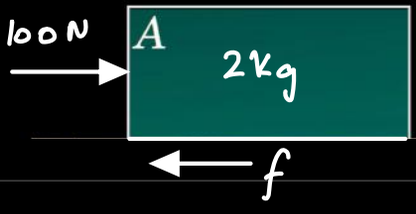
Solution: Once again let us assume static friction. Then:

$a_A = a_B = (100-20)/(2+1) = 80/3\ ms^{-2}$ →

Now considering FBD of A alone:

$100 - f = 2 \times 80/3 = 160/3$

$\Rightarrow f = 100 - \frac{160}{3} = \frac{140}{3}$ N, $f_l = 20 \times 0.5 = 10$N



Now $f > f_l$ thus our assumption has failed.

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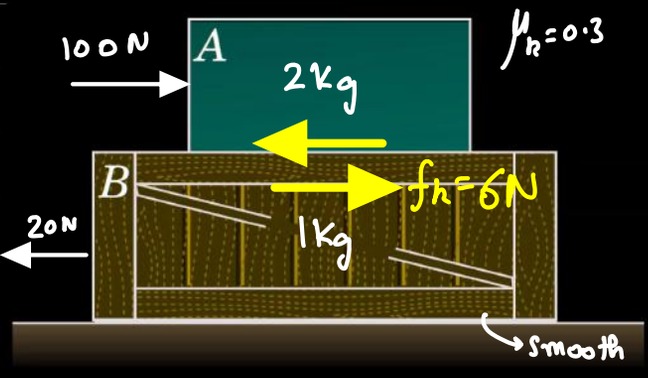
Now we need to solve the problem again assuming kinetic friction. $f_k = 20 \times 0.3 = 6N$

Thus:

$a_A = (100 - 6)/2 = 47 m/s^2 \longrightarrow$ Ans

$a_B = (20 - 6)/1 = 14 m/s^2 \longleftarrow$ Ans

and $f = f_k = 6N$ Ans.



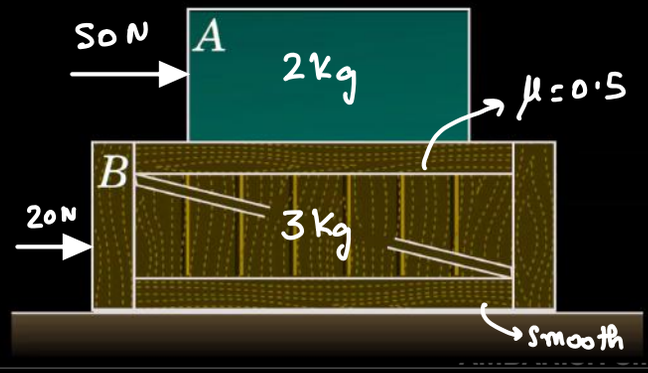
→ While with static friction it does not matter even if you choose the wrong direction, with kinetic friction, the friction must be applied in the correct direction which can be found by assuming frictionless surfaces.

METHOD 2: ASSUMING KINETIC FRICTION:

1. Assume kinetic friction and apply it in correct direction
2. Find accelerations of the blocks under this assumption
3. If the relative velocity at any surface is along the kinetic friction, our assumption has failed and we need to solve again assuming static friction. Otherwise if the relative velocity is against the frictional forces, our assumption is correct and we already have the correct solution.

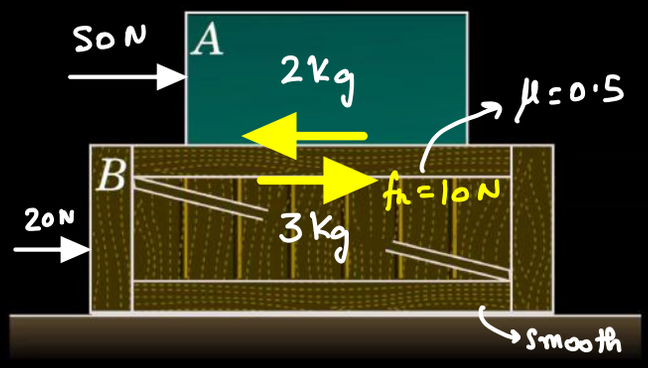
→ This method must be used only if $\mu_s = \mu_k$ otherwise we may reach wrong conclusions. Relative slipping will start only if required friction exceeds limiting friction. Thus kinetic friction values that we assumed will practically never be achieved in the first place if required friction for moving together does not exceed limiting friction.

EXAMPLE: In the shown system, $\mu_s = \mu_k = \mu = 0.5$. Find accelerations of the blocks and friction between them.



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Solution: Assuming kinetic friction,



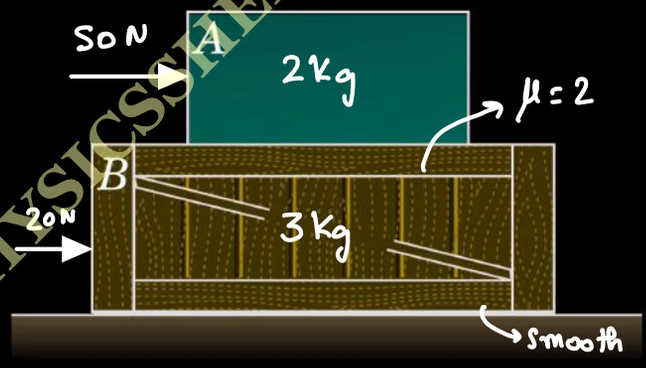
$a_A = (50-10)/2 = 20\,m/s^2 \rightarrow$

$a_B = (20+10)/3 = 10\,m/s^2 \rightarrow$

Now $a_{AB} = a_A - a_B = 10\,m/s^2 \rightarrow$

We see that friction is opposite to the relative acceleration, thus our assumption is correct thus $a_A = 20\,m/s^2 \rightarrow$ Ans. $a_B = 10\,m/s^2 \rightarrow$ Ans and $f = f_k = 10\,N$ Ans

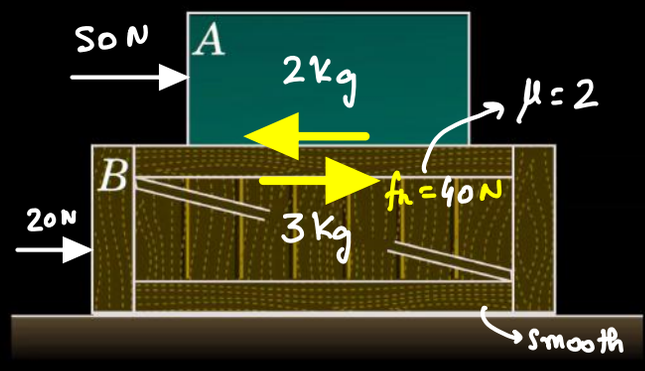
EXAMPLE: For the shown system find the accelerations of the blocks and friction between them.



Solution: Assuming kinetic friction:

$a_A = (50-40)/2 = 5\,m/s^2 \rightarrow$

$a_B = (20+40)/3 = 20\,m/s^2 \rightarrow$



Thus $a_{AB} = a_A - a_B = 15\,m/s^2 \leftarrow$

Here we see that relative acceleration of A and frictional force on it are in same direction. Thus our assumption has failed and we need to solve again using static friction assumption.

Now, assuming static friction:

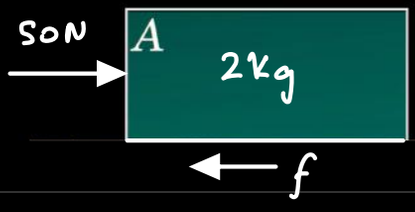
$a_A = a_B = (50+20)/(3+2) = 14\,m/s^2 \rightarrow$

Now considering FBD of 2Kg alone,

$50 - f = 2 \times 14 = 28 \Rightarrow f = 50-28 = 22$

thus $f = 22\,N$ Ans.

$a_A = a_B = 14\,m/s^2 \rightarrow$ Ans.

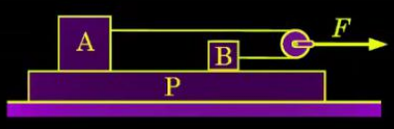


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Handling Problems With Friction At Multiple Interfaces With Unknown Condition Of Impending Motion Or Sliding:

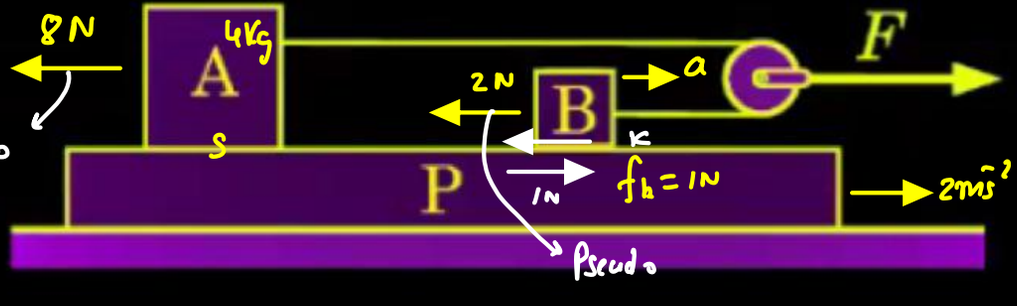
If there are frictional forces at multiple interfaces, the possibility of cases increases exponentially. For instance with two rough surfaces, there are 2^2 possibilities (S,S), (S,K), (K,S), (K,K) K: Kinetic S: Static. Similarly with friction at 3 interfaces, there are 2^3 possibilities. For handling such problems, we generally start with an intuitive guess for the first case and if we get contradiction at any interface, we toggle the friction (i.e S → k or k → s) at that surface to arrive at a new case and check for its validity according to methods described in previous sections. (i.e under static assumption required friction should not exceed limiting value and under kinetic assumption, relative acceleration should not be along friction).

Example: Pathfinder: Build36:



In the setup shown, blocks A and B of masses 4.0 kg and 1.0 kg are placed on a platform P of mass 1.0 kg that is placed on a horizontal frictionless floor. The blocks are connected by a light inextensible string that passes round an ideal pulley. Coefficients of static and kinetic friction between the blocks and the platform are 0.16 and 0.10. On applying a horizontal force F on the pulley, acceleration of the platform becomes 2.0 m/s². Find magnitude of the force F, frictional forces on the blocks and their accelerations. Acceleration due to gravity is 10 m/s².

Solution: Thought process: SS looks unlikely (will make the problem too easy). Now 4Kg block is the heavier one and thus causes Normal Reaction and limiting friction so it is likely to have static friction and 1Kg is the lighter one and has smaller limiting friction so it is likely to have kinetic friction. So we start with case S-k for A-B respectively.



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With this assumption, F must be greater than $(m_A+m_P) \times 2 ms^{-2}$ i.e $F > 10N$, thus relative acceleration of B must be forward. Now limiting friction for A, $f_{s_A} = 40 \times 0.16 = 6.4 N$.

Now applying $\Sigma F_{ext} = \Sigma m_i a_i$:

$$F = (4+1) \times 2 + 1 \times (2+a) \Rightarrow F = 12 + a \quad (1)$$

Now considering FBD of A and P combined from ground frame:

$$\frac{F}{2} + 1 = (4+1) \times 2 \quad (2) \Rightarrow F = 18N, \; a = 6 ms^{-2}$$

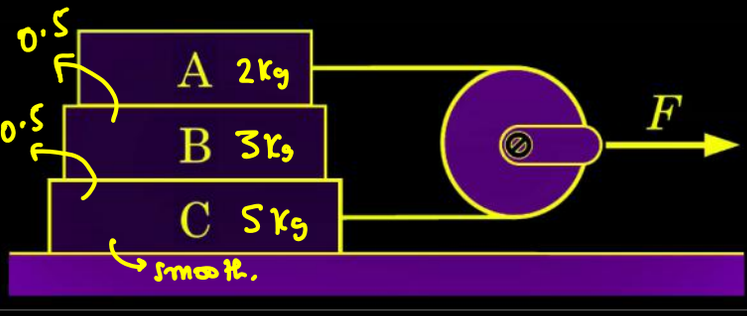
Now required f on A: $\frac{F}{2} - f_A = 4 \times 2 \Rightarrow f_A = 9 - 8 = 1N$.

this is less than limiting friction of 6.4 N. Thus this case seems plausible, and we can be sure if we can eliminate S-S case.

With S-S case, $F = (4+1+1) \times 2 = 12N$, Now required value of friction of B: $\frac{12}{2} - f_B = 1 \times 2 \Rightarrow f_B = 4N$ which is more than f_{l_B} (i.e $10 \times 0.16 = 1.6N$). Thus S-S case is indeed ruled out thus our required solution:

$F = 18N$, $a_A = 2 ms^{-2}$, $a_B = 2+6 = 8 ms^{-2}$, $f_B = 1N$, $f_A = 1N$ Ans.

EXAMPLE: In the shown system, the floor is smooth and coefficient of friction between the blocks, $\mu_s = \mu_k = 0.5$. Find (a) maximum value of F so that all blocks may move together (b) Acceleration of each block and frictional forces if F is double of the value in (a) part.



Solution: Limiting friction $f_{s_{AB}} = 20 \times 0.5 = 10N$, $f_{s_{BC}} = 50 \times 0.5 = 25N$.

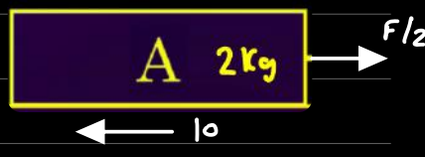
Thought process: With maximum value of F, atleast one of the frictional forces must reach its limiting value, now since limiting friction at A is smaller and it is a lighter block we expect it

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to be at the impending motion (on limiting friction f_{AB}).

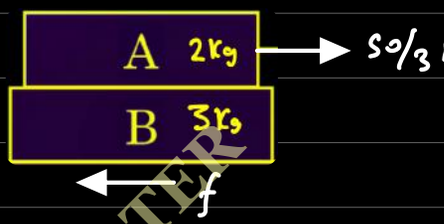
Now combined acceleration: $a = F/10 \rightarrow$ ①

Now considering impending slipping at AB interface,



$\frac{F}{2} - 10 = 2a$ ②, from ① & ②:

$F = 100/3$ N, $a = 10/3$ m/s^2 →, Now let us check the required friction at BC interface: Considering combined FBD of A and B, $50/3 - f = 5 \times 10/3$



$\Rightarrow f = 0$ N, $f_{l_{BC}} = 50 \times 0.5 = 25$ N, thus required value is less than limiting value so we have the required case. Thus:

$F_{max} = 100/3$ N Ans. (a)

Now when F is double this value we expect either (s_{BC}, k_{AB}) or (k_{BC}, k_{AB}) cases. Let us try out kinetic friction at both interfaces.

Now $a_A = \left(\frac{100}{3} - 10\right)/2 = 35/3$ m/s^2 →

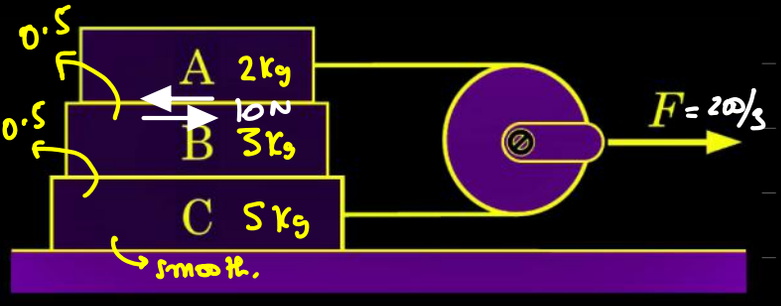


$a_B = 35/3$ m/s^2 →

$a_C = \left(\frac{100}{3} - 25\right)/5 = \frac{5}{3}$ m/s^2 →

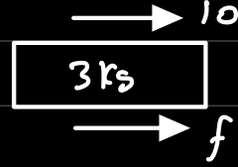
$a_{BC} = \frac{35}{3} - \frac{5}{3} = 10$ m/s^2 →, but friction of B is also forward i.e along the direction of relative acceleration. Thus there is contradiction. So we now try out the second case i.e (k_{AB}, s_{BC}):

Now $a_A = (100/3 - 10)/2 = 35/3$ m/s^2 →



$a_B = a_C = (100/3 + 10)/8 = 65/12$ m/s^2 →

Now let us find required friction at BC interface. Considering FBD of 3kg block alone, $10 + f = 3 \times \frac{65}{12} \Rightarrow f = 6.25$ N



But limiting friction $f_{l_{BC}} = 25$ N thus $f < f_{l_{BC}}$ thus,

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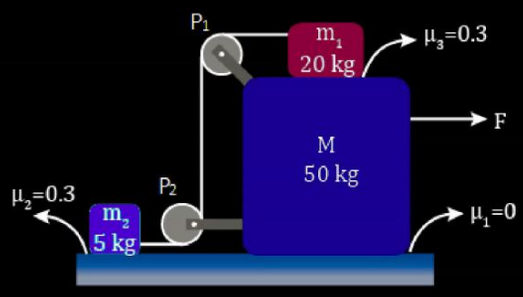
now we have the required case. Thus: $a_A = 35/3\ m/s^2$ → Ans
$a_B = a_C = 65/12\ m/s^2$ → Ans $f_{AB} = 10N$, $f_{BC} = 6.25N$ Ans.

EXAMPLE: JEE: 2000:

In the figure masses m_1, m_2 and M are 20 kg, 5 kg and 50 kg respectively. The coefficient of friction between M and ground is zero. The coefficient of friction between m_1 and M and between m_2 and ground is 0.3. The pulleys and the strings are massless. The string is perfectly horizontal between P_1 and m_1 and also between P_2 and m_2. The string is perfectly vertical between P_1 and P_2. An external horizontal force F is applied to the mass M. [Take g=10 m/s².]

(a) Draw a free body diagram of mass M, clearly showing all the forces.

(b)Let the magnitude of the force of friction between m_1 and M be f_1 and that between m_2 and ground be f_2. For a particular force F, it is found that $f_1 = 2f_2$. Find f_1 and f_2. Write equations of motion of all the masses. Find F, tension in the string and acceleration of the masses.

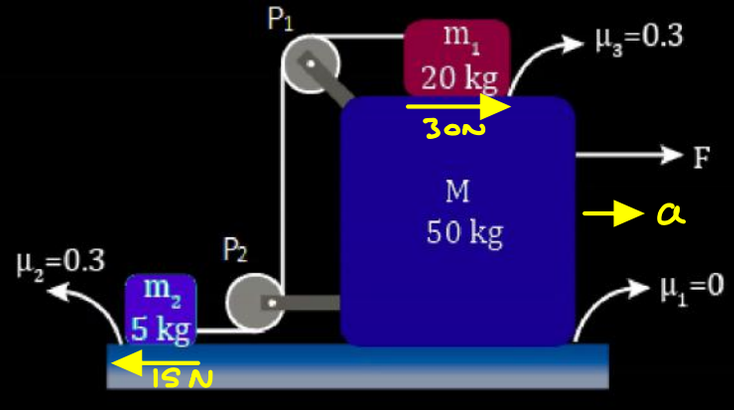


Solution: limiting friction between m_1 and M, $f_{l_1} = 200 \times 0.3 = 60N$
limiting friction between m_2 and Ground, $f_{l_2} = 50 \times 0.3 = 15N$.
Now $f_1 = 2f_2$, this means that friction between m_1 and M can't be limiting because otherwise if $f_1 = 60N$, $f_2 = 30N$ which is more than limiting value of f_2 (i.e 15N). So that leaves us with two possibilities, (S, S) or (k_{m_2}, S_{m_1}). Now if both frictions are static then entire system will have to be at rest and considering equilibrium of m_1 and m_2, $f_1 = f_2$ (∵ both are pulled by same tension) however this is also ruled out since it is given that $f_1 = 2f_2$, that leaves with only possibility (k_{m_2}, S_{m_1}). In that case, $f_2 = 15N$, $f_1 = 30N$, and all three blocks must move at same acceleration due to constraint.

Now considering combined FBD of m_2 and m_1 and making equation in horizontal direction:

$(30 - 15) = (20 + 5)a$

$\Rightarrow a = \frac{15}{25} = \frac{3}{5}\ m/s^2$ Ans.



Now considering combined FBD of all the blocks together:

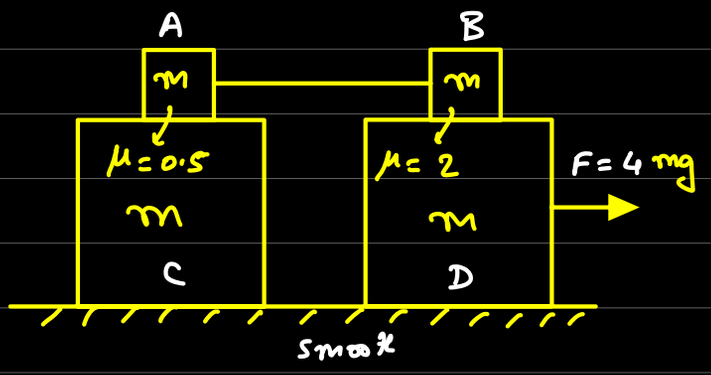
$F - 15 = (50 + 20 + 5) \times \frac{3}{5} \Rightarrow F = 60N$ Ans. Also $T - 15 = 5 \times \frac{3}{5} \Rightarrow T = 18N$ Ans.

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EXAMPLE:

Consider the system shown in the figure. Find the frictional forces and accelerations of the blocks



Solution: Let us start with assumption $(S_{AC}, S_{BD}) \Rightarrow a = 4mg/4 = g \rightarrow$

However limiting friction on C is $0.5\,mg$ so it can't have an acceleration greater than $0.5g$. So our assumption is wrong. So let us try (K_{AC}, S_{BD}). Now limiting friction $f_{AC} = 0.5mg$. Thus considering combined fBD of ABD,

$a_A = a_B = a_D = (4mg - 0.5mg)/3m = 7g/6 \rightarrow$

Now let f_{BD} be the required friction between B & D for this case. Considering combined FBD of AB, $f - 0.5mg = 2m\,(7g/6)$

$\Rightarrow f = 17/6\,mg$, However limiting friction, $f_{s_{BD}} = 2mg$, so once again our assumption failed. Now let us try the possibility, (S_{AC}, K_{BD}) Considering combined FBD of ABC under this assumption:

$a_A = a_B = a_C = (2mg/3m) = 2g/3$. Now let f_{AC} be required friction under this assumption. Then considering fBD of C alone:

$f = m \times 2g/3$ i.e $f = 2/3\,mg$, but limiting value is $0.5mg$, so this assumption also failed. Now let us try final possibility (K_{AC}, K_{BD}). Now considering combined FBD of A and B,

$a_A = a_B = (2mg - 0.5mg)/2m = 3g/4 \rightarrow$

$a_D = (4mg - 2mg)/m = 2g \rightarrow$, so D has a rightward acc^n wrt B, so no violation of kinetic friction condition.

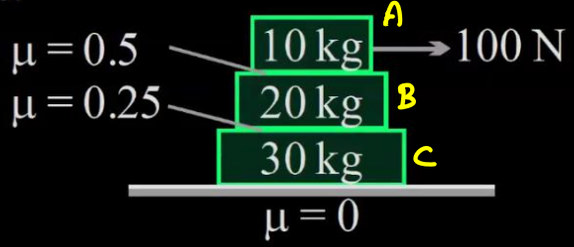
Now $a_C = (0.5mg)/m = g/2$, thus A has rightward acc^n wrt C, so no violation here as well. Thus finally:

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$a_A = a_B = 3g/4$, $a_C = g/2$, $a_D = 2g$ Ans. , Now T can be calculated considering the system AC, $T = m \times 3g/4 + m \times g/2 = \frac{5}{4} mg$ Ans.

EXAMPLE:

Three blocks are kept as shown in the figure. Acceleration of 20 kg block with respect to ground is:



(A) 5 m/s² (B) 2 m/s² (C) 1 m/s² (D) none of these

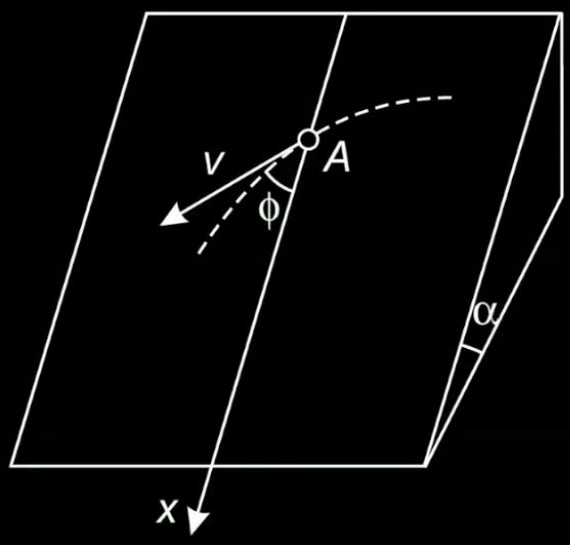
Solution: Assuming (f_{AB}, f_{BC}), $a_A = (100-50)/10 = 5 m/s^2 \rightarrow$

$a_B = (50 - 300 \times 0.25)/20$, this is negative i.e leftwards, ∴ not possible. So we now try (f_{AB}, f_{BC}), thus with this assumption, $a_B = a_C = 50/50 = 1 m/s^2 \rightarrow$ and $a_A = 5 m/s^2 \rightarrow$

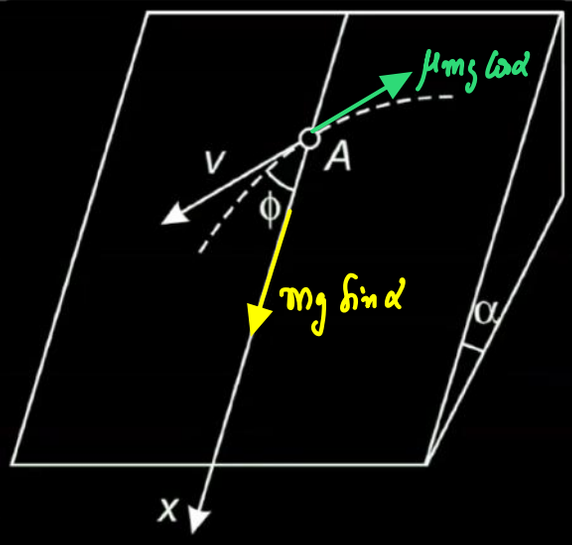
Now if f is required friction between B and C, $f = 30 \times 1 = 30 N$ while limiting friction here is 300×0.25 i.e $75 N$, thus $f < f_s$ and hence we have the right case. Thus $a_B = 1 m/s^2 \rightarrow$ Ans.

EXAMPLE: IRODOV: 1.106:

A small disc A is placed on an inclined plane forming an angle α with the horizontal (Fig. 1.27) and is imparted an initial velocity v_0. Find how the velocity of the disc depends on the angle ϕ if the friction coefficient $k = \tan\alpha$ and at the initial moment $\phi_0 = \frac{\pi}{2}$.



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Solution: Applying second law along tansent:

$$mg\sin\alpha\cos\phi - \mu mg\cos\alpha = m\,dv/dt \quad (1)$$

Applying second law along steepest descent:

$$mg\sin\alpha - \mu mg\cos\alpha\cos\phi = m\,dv_x/dt \quad (2)$$

and $\mu = \tan\alpha$

(2)/(1) yields:

$$\frac{dv_x}{dv} = -1$$

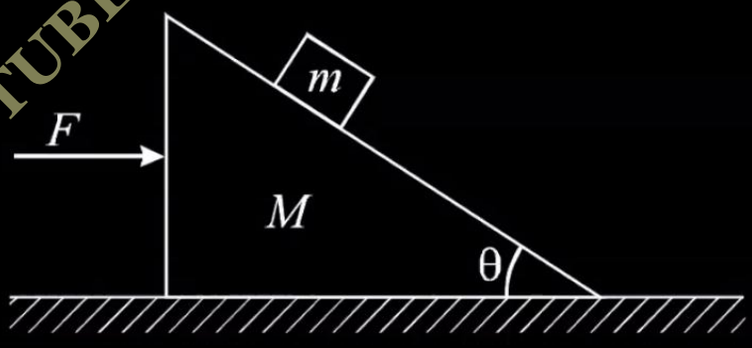
$\Rightarrow v_x = -v + C$ but at $t=0$, $v_x = 0$

and $v = v_0 \Rightarrow v_x = -v + v_0 = v\cos\phi \Rightarrow v(1+\cos\phi) = v_0$

$$\Rightarrow v = \frac{v_0}{1+\cos\phi} \text{ Ans.}$$

EXAMPLE:

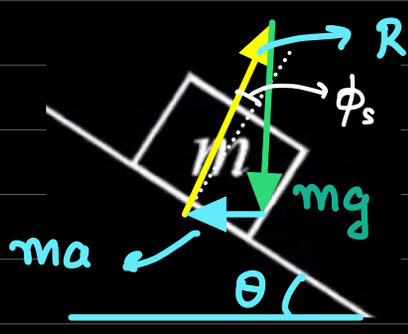
A bar of mass m is placed on a triangular block of mass M as shown in figure. The friction coefficient between the two surface is μ and ground is smooth. Find the minimum and maximum horizontal force F required to be applied on block so that the bar will not slip on the inclined surface of block.



Solution: If the force is very small, block will have downward slipping tendency due to small component of pseudo force. The block will be in impending relative motion and the three forces on block will be: mg, contact force R and pseudo force. Now making the force triangle and using sine rule:

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$$\frac{mg}{\sin(90^\circ-\theta+\phi_s)} = \frac{ma}{\sin(\theta-\phi_s)} = \frac{R}{\sin 90^\circ}$$



$$\Rightarrow \quad a = \frac{\sin(\theta-\phi_s)}{\cos(\theta-\phi_s)}\, g = g\tan(\theta-\phi_s)$$

Now considering combined FBD :

$$F_{min} = (M+m)\,a = (M+m)\,g\tan(\theta-\phi_s)$$

$$\Rightarrow F_{min} : (M+m)\,g\left[\frac{\tan\theta-\tan\phi_s}{1+\tan\theta\tan\phi_s}\right] = (M+m)\,g\left(\frac{\tan\theta-\mu}{1+\mu\tan\theta}\right)$$

If $\tan\theta < \mu \Rightarrow$ angle of incline is less than angle of repose and the block will remain at rest without any application. (Infact a negative force in limiting equilibrium means that instead of pushing the system will have to be pulled to attain limiting equilibrium condition).

Now when F is too large, the limiting friction will act down the plane instead of up the plane. We could do the entire analysis again, however working smartly we can model this situation by simply replacing μ with $-\mu$. Thus :

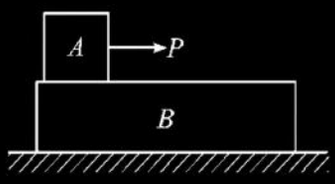
$$F_{max} = (M+m)\,g\left[\frac{\tan\theta+\mu}{1-\mu\tan\theta}\right] = (M+m)\,g\tan(\theta+\phi_s)$$

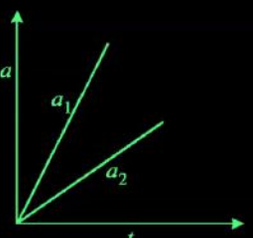
When $\theta+\phi_s = 90^\circ$, $F_{max} \to \infty$, When $\theta+\phi_s > 90^\circ$, still allowable max force is infinity and limiting equilibrium is never achieved (equilibrium is achieved with a value of friction less than limiting value for instance if $\theta = 90^\circ$, Normal reaction can go up to infinity however required frictional force is only mg.

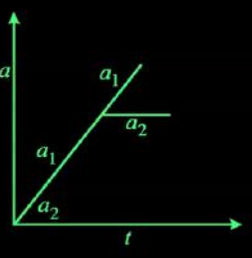
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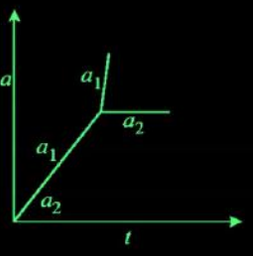
EXAMPLE:

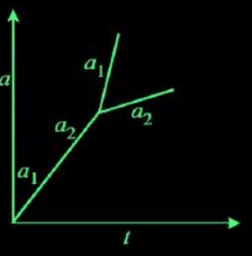
Block A is placed on block B, whose mass is greater than that of A. There is friction between the blocks, while the ground is smooth. A horizontal force P, increasing linearly with time, begins to act on A. The acceleration a_1 and a_2 of A and B respectively are plotted against time (t). Choose the correct graph :



(A)


(B)


(C)


(D)


Solution: Initially Both blocks must accelerate together until impending slipping and after that acceleration of B must become constant while acceleration of A must be increasing at a faster rate because earlier frictional force was increasing in proportion to applied force, however now frictional force becomes constant (after slip starts) and same increase in external force must produce larger increase in acceleration. Thus required graph is (C).

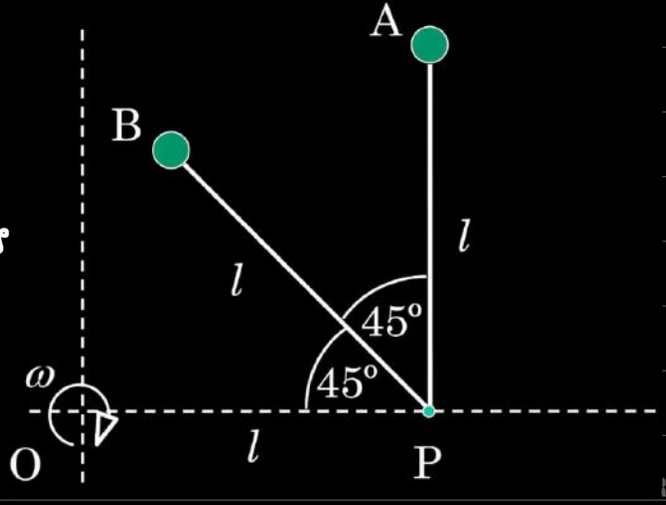
Alternatively, $F = m_1 a_1 + m_2 a_2 \Rightarrow \frac{dF}{dt} = m_1 \frac{da_1}{dt} + m_2 \frac{da_2}{dt}$

$\Rightarrow \frac{da_1}{dt} = \frac{1}{m_1}\left[\frac{dF}{dt} - m_2 \frac{da_2}{dt}\right]$, after kinetic friction, $\frac{da_2}{dt} = 0$

thus slope of $a_1 - t$ graph increases.

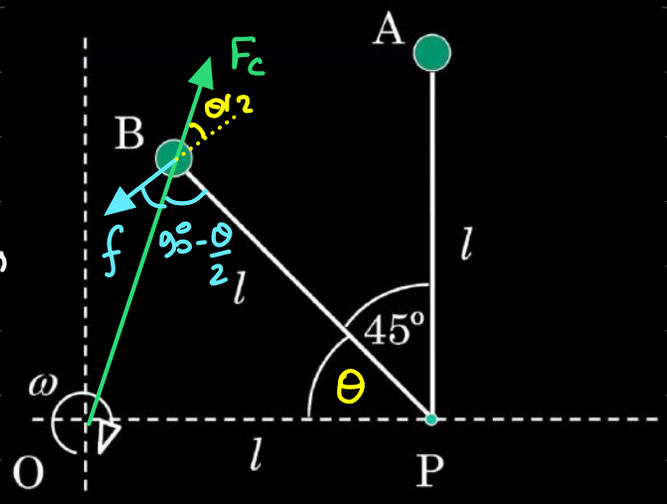
EXAMPLE:

Two small identical discs A and B are tied to a nail P on a large horizontal platform with help of light strings of length l each. The platform starts rotating with gradually increasing angular speed ω. For what value of ω do the discs A and B begin to slip if coefficient of friction is μ.



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Solution: We will solve the problem for a general θ as shown in the figure. In the limiting equilibrium, component of centrifugal force perpendicular to PB will balance the frictional force.



Thus

$$m\omega^2 \times 2l\sin(\theta/2)\cos(\theta/2) = \mu mg \Rightarrow \omega = \sqrt{\frac{\mu g}{l\sin\theta}}$$

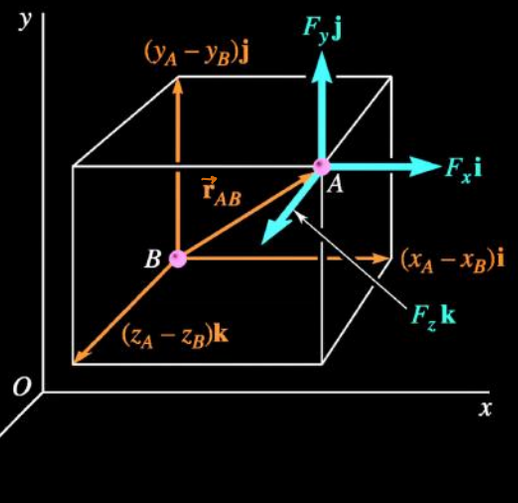
Now we can just put $\theta = 45°$ for B and $\theta = 90°$ for A. (Ball A will begin to slip before ball B).

TORQUE BALANCE: If an object is at rest in some frame then net torque of all the forces acting on the object about every point of that frame is zero (if the frame is non-inertial, we also need to take the torques due to pseudo forces). We will prove this result in the chapter on rotational mechanics. Here we will just be using the result without proof. We introduced the idea of torque in our study of vectors, the following section is a brief recap of the idea of torque.

TORQUE (MOMENT) OF A FORCE:

Let a force $\vec{F}$ act at a point A whose position vector wrt a point B is $\vec{r}_{AB}$ then we define torque of the force about point B as:

$$\boxed{\vec{\tau} = \vec{r}_{AB} \times \vec{F}}$$



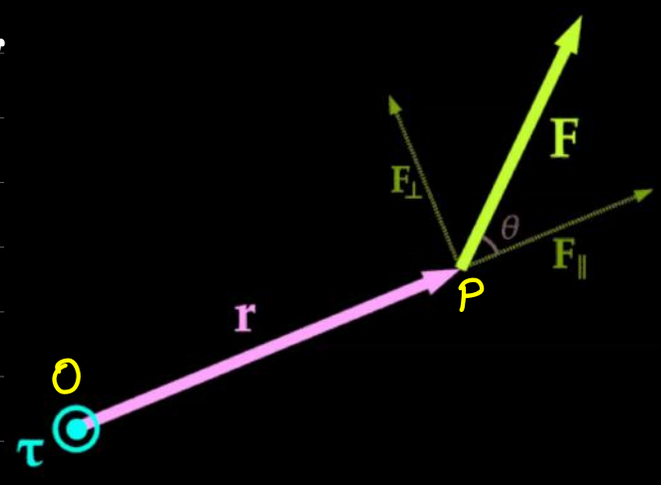
→ Here $\vec{r}_{AB}$ is position of A wrt B i.e

$$\vec{r}_{AB} = \vec{r}_A - \vec{r}_B = (x_A - x_B)\hat{i} + (y_A - y_B)\hat{j} + (z_A - z_B)\hat{k}.$$

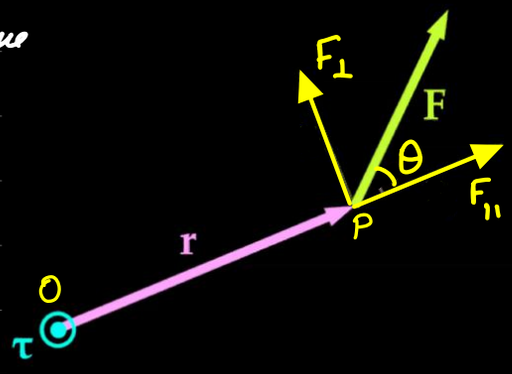
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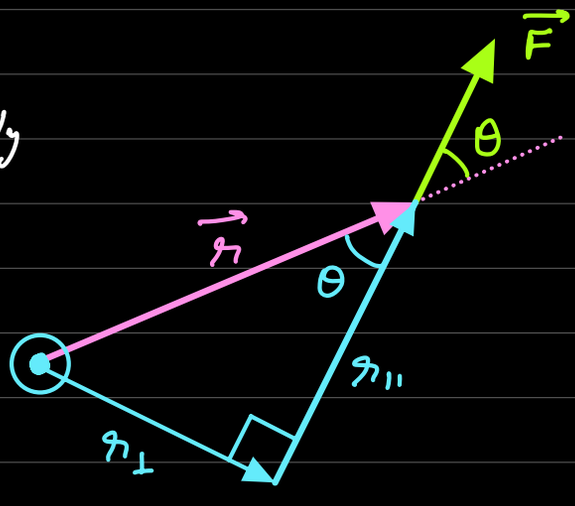
Torque As A Signed Scalar:

Even though torque is a vector, for problems involving forces only in two dimensions, we may treat it as a signed scalar. In the plane of the page, torque of any force about any point may be considered as either clockwise or anti-clockwise. (We can see this by imagining a stick pivoted at point about which we are finding the torque with other end at point of application of force. Now visualise the effect of the force on this pivoted stick). For instance, in the figure shown above, the torque of $\vec{F}$ about O is anticlockwise as can be seen by imagining the stick OP (pink). If clockwise torques are assumed to be positive then, anti-clockwise torques must be considered negative and vice versa.



Magnitude Of Torque: Recall that the torque $\vec{\tau} = \vec{r} \times \vec{F}$ thus using definition of cross product $\tau = rF\sin\theta$, now if we resolve F in to components $\parallel$ and $\perp$ to r we recognize that $F\sin\theta$ is nothing but component of $\vec{F}$ $\perp$ to r. Thus we can also write $\boxed{\tau = r F_\perp}$ or simply "distance times perpendicular force." Alternatively we could also resolve $\vec{r}$ in to components $\parallel$ and $\perp$ to F and we recognize that $r_\perp = r\sin\theta$, thus $\tau = rF\sin\theta = F r\sin\theta$ $= F r_\perp$ thus another way of writing torque is $\boxed{\tau = F r_\perp}$ or simply,



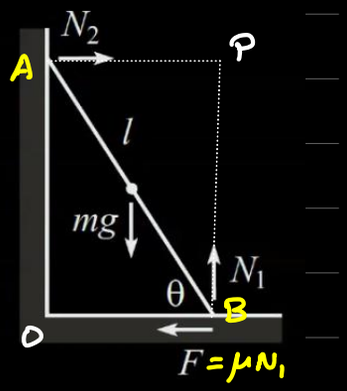


Torque is Force times perpendicular distance.

With these definitions now we can balance the torques on any still object (keep in mind the sign of torques).

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Example (Leaning ladder): A ladder leans against a frictionless wall. If the coefficient of friction with the ground is μ, what is the smallest angle the ladder can make with the ground and not slip?



Solution: By vertical force balance: $N_1 = mg$ ①

By torque balance about point P: $\mu N_1 \times l \sin\theta = mg \times \frac{l}{2}\cos\theta$ ②

$\Rightarrow \mu mg \sin\theta = \frac{mg}{2}\cos\theta \Rightarrow \tan\theta = 1/2\mu$. Thus minimum possible angle θ is $\tan^{-1}(1/2\mu)$. (We took $f=\mu N_1$ because of impending slipping in the limiting case).

→ Note that as θ decrease, torque arm of mg increases while that of friction decreases, thus if θ is reduced below this value, anti-clockwise torque of mg will dominate the clockwise torque of friction causing an angular acceleration of the ladder.

CONCURRENCY THEOREM: If a body is in equilibrium under action of three forces then the three forces must be concurrent (or parallel i.e concurrence at ∞). This can be proven very easily. Suppose lines of actions of two forces intersect at some point, then torque of those two forces will be zero about that point. Now if third force does not pass through that point then its torque will be non zero thus producing a non-zero resultant torque.

EXAMPLE:

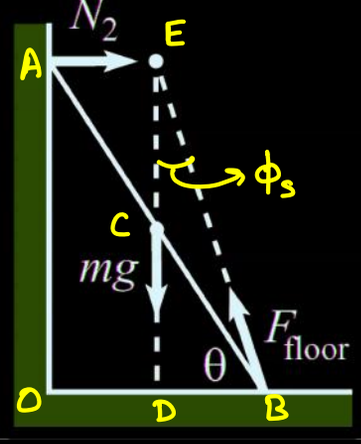
Solve the ladder problem using concurrency theorem.

Solution: Consider the adjacent figure. Here:

$BD = BC\cos\theta = ED\tan\phi_s$ thus:

$\frac{l}{2}\cos\theta = l\sin\theta \times \tan\phi_s \Rightarrow \tan\theta = \frac{1}{2\tan\phi_s}$

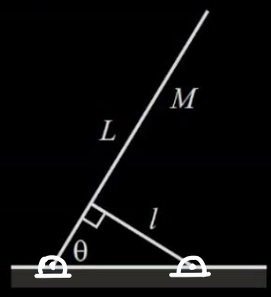
$\Rightarrow \tan\theta = 1/2\mu$ for limiting equilibrium.



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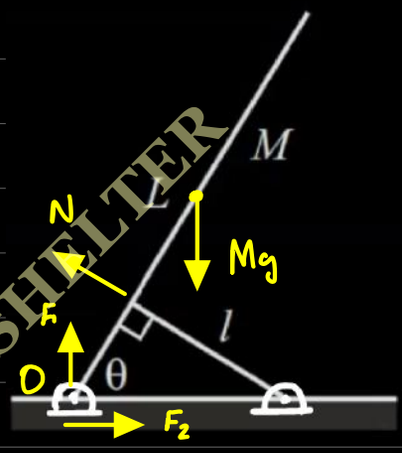
EXAMPLE: (MORIN):

A ladder of length L and mass M has its bottom end attached to the ground by a pivot. It makes an angle θ with the horizontal and is held up by a massless stick of length ℓ that is also attached to the ground by a pivot (see Figure). The ladder and the stick are perpendicular to each other. Find the force that the stick exerts on the ladder.



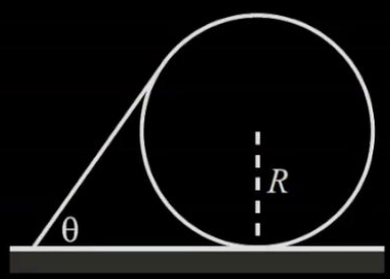
Solution: Consider the FBD of the massive stick as shown in the figure. Balancing torques about point O, $N\ell \cot\theta = Mg\frac{L}{2}\cos\theta$

$\Rightarrow N = \frac{L}{2\ell} Mg \sin\theta$ Ans.



EXAMPLE: MORIN:

A stick of mass density per unit length ρ rests on a circle of radius R (see Figure). The stick makes an angle θ with the horizontal and is tangent to the circle at its upper end. Friction exists at all points of contact, and assume that it is large enough to keep the system at rest. Find the friction force between the ground and the circle.



Solution: The circle is in equilibrium under action of three forces, i.e. two contact forces and the weight, thus all three forces must be concurrent at the lowest point of the circle.

To:

$$\rho\ell g \times \frac{\ell}{2}\cos\theta = F_1 \cos(\theta/2)\,\ell \quad (1)$$

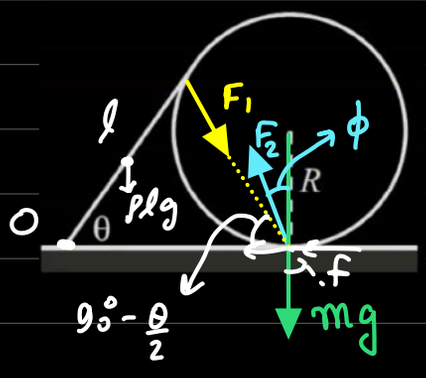
$$f = F_1 \cos(90^\circ - \theta/2) \quad (2)$$

(2)/(1)

$$f = \frac{\rho\ell g}{2}\cos\theta \sec\theta/2 \sin(\theta/2) = \frac{\rho\ell g}{2}\tan(\theta/2)\cos\theta$$

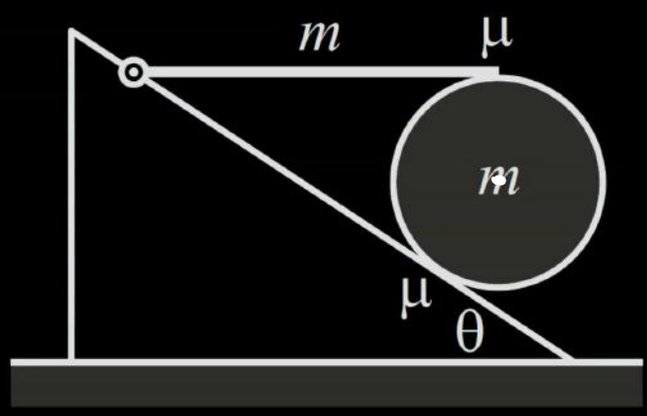
and $\ell = R\cot(\theta/2)$

$$\Rightarrow f = \frac{\rho R g \cos\theta}{2}$$ Ans.

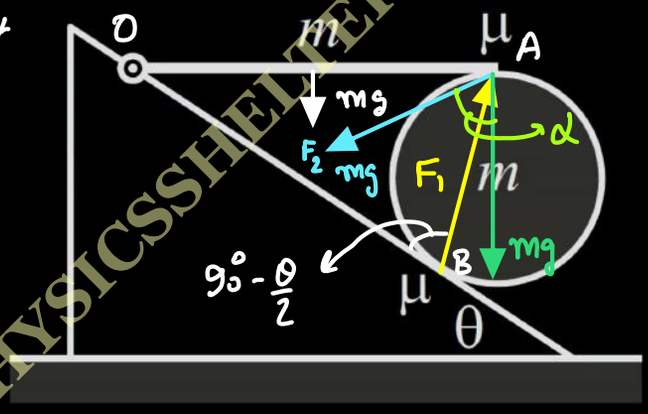


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EXAMPLE: A horizontal stick has its left end pivoted on an incline of angle θ, while the right end rests on top of a cylinder, as shown in the figure. Find (a) Normal reaction between the incline and the cylinder (b) Minimum μ for which the system does not slip anywhere.



Solution: Consider the adjacent diagram. The reaction force at B must be concurrent with intersection of mg and reaction force at B. Now balancing torque about point O for the combined system:



$$mg\frac{l}{2} + mgl - N_B l = 0$$

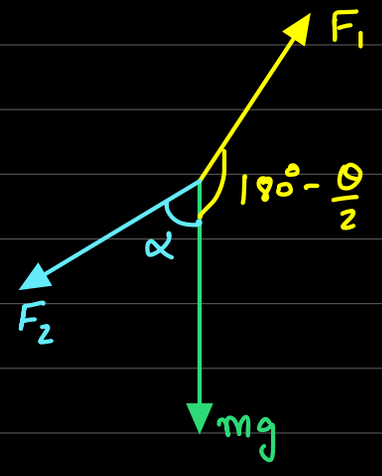
where l is the length of the stick and N_B is the normal reaction at B. Thus:

$$N_B = 3/2\ mg \quad \text{Ans (a)}. \quad \text{Now } N_B = F_1 \cos\theta/2 \Rightarrow F_1 = \frac{3}{2}mg\sec(\theta/2) \quad (A)$$

Now F_1 must lie within the cone of friction, thus $\frac{\theta}{2} < \tan^{-1}\mu$ (1)

$\Rightarrow \mu > \tan(\theta/2)$ (1) (Considering no slipping at B).

Now Applying Lami's theorem for the cylinder:



$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin(180° - \theta/2)} = \frac{mg}{\sin(180° + \theta/2 - \alpha)}$$

$$\Rightarrow \frac{\sin(\alpha - \theta/2)}{\sin\alpha} = \frac{mg}{F_1} = \frac{2}{3}\cos(\theta/2)$$

$$\Rightarrow \frac{\sin\alpha\cos(\theta/2) - \cos\alpha\sin(\theta/2)}{\sin\alpha} = \frac{2}{3}\cos(\theta/2)$$

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$$\Rightarrow \quad \cos(\theta/2) - \cot\alpha \sin(\theta/2) = \frac{2}{3}\cos(\theta/2)$$

$$\Rightarrow \quad \cot\alpha = \frac{1}{3}\cot(\theta/2) \quad \Rightarrow \quad \tan\alpha = 3\tan(\theta/2)$$

Also, $\alpha < \tan^{-1}\mu \quad \Rightarrow \quad \mu > 3\tan(\theta/2)$ (2)

Strictest bound is obtained from (2), thus $\mu > 3\tan(\theta/2)$ ans.

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